CALCULUS

Chapter 15

Integral Calculus

15

SYLLABUS 5.5 5.9 5.10 5.11 5.15 5.16 5.17

Your car's speedometer tells you how fast you are going at every instant. Suppose you record it for an entire journey. Can you work out how far you travelled, using only the speed?

Over a tiny slice of time, your speed barely changes, so the distance covered in that slice is roughly speed times the slice of time. Add up all the slices and you have the total distance. As the slices shrink toward zero width, the rough sum becomes exact. What you are computing is the area under the speed-time graph.

This is the second great idea of calculus, and it arrives as two ideas rather than one. The first is **reversing a derivative**: given a rate of change, recover the quantity it came from. The second is **accumulating**: adding up a great many thin slices to get a total. The area under a graph is exactly this kind of total.

They have no obvious connection. Reversing a derivative is a problem in algebra. Adding slices is a problem in geometry. Yet the car journey links them, because the distance travelled is both the reverse of the velocity and the area under the velocity graph.

The chapter is built in that order. Section 15.1 does the reversal on its own, with no mention of area. Section 15.2 defines the area on its own, with no mention of derivatives, and then proves that the two are the same calculation. That result is the Fundamental Theorem of Calculus, and once it is in place the accumulated totals, areas between curves, volumes of revolution and distances travelled all become routine.

15.1 Antiderivatives

This chapter reverses differentiation. Given a function f, the task is to find a function whose derivative is f. Such a function is called an **antiderivative** of f.

An antiderivative is never unique. The derivative of a constant is zero, so any constant may be added to an antiderivative without altering its derivative. Antiderivatives therefore occur in families whose members differ by a constant.

DEF F is an **antiderivative** of f when $F'(x) = f(x)$. If F is one antiderivative, then $F(x) + C$ is an antiderivative for every constant C, and these are all of them.

Reversing the power rule means raising the power by one and dividing by the new power.

KEY · IB An antiderivative of x^n is $\dfrac{x^{n+1}}{n+1} + C$, for $n \neq -1$.

The exception $n = -1$ is handled separately in the next section. Constant multiples and sums are reversed term by term, just as they are differentiated term by term.

EXAMPLE 15.1

Find the antiderivatives of $3x^2 - 4x + 1$.

Reverse each term, raising each power by one:

$$F(x) = x^3 - 2x^2 + x + C.$$

Check by differentiating: $\frac{d}{dx}(x^3 - 2x^2 + x + C) = 3x^2 - 4x + 1$. ✓ Differentiating the answer verifies it completely.

Naming the whole family in words each time is cumbersome, so it is given a symbol of its own.

KEY

$$\int f(x)\,dx = F(x) + C \qquad \text{means} \qquad F'(x) = f(x).$$

This is the **indefinite integral** of f, and C is the **constant of integration**. It is easily omitted, and an answer written without it names a single antiderivative rather than the whole family.

15.1.1 Boundary conditions

The $+C$ means that an antiderivative is not determined until further information is supplied. A single condition, such as a point through which the curve passes, fixes C and selects one member of the family.

EXAMPLE 15.2

A curve has gradient $\frac{dy}{dx} = 6x^2 - 2$ and passes through $(1, 4)$. Find its equation.
Antidifferentiate to recover y:

$$y = 2x^3 - 2x + C.$$

Use the point $(1, 4)$: $4 = 2 - 2 + C \implies C = 4$. So $y = 2x^3 - 2x + 4$.

YOUR TURN 15.1 Antiderivatives

1. Find the antiderivatives of each function.

(a) $4x^3 - 6x$
(b) $x^2 + 3x - 5$
(c) $\frac{6}{x^3}$
(d) $\sqrt{x}$
(e) $x^2 - \frac{1}{x^2}$
(f) $\frac{x^3 - 2x}{x}$
(g) $x^{-1/2}$
(h) $(2x)^3$

2.

(a) A curve has $\frac{dy}{dx} = 4x - 3$ and passes through $(2, 5)$. Find y.
(b) A curve has $\frac{dy}{dx} = 3\sqrt{x}$ and passes through $(4, 20)$. Find y.
(c) A curve has $\frac{d^2y}{dx^2} = 6x$, with $\frac{dy}{dx} = 1$ and $y = 2$ when $x = 0$. Find y.
(d) Explain why a gradient function alone does not determine a curve, and what extra information does.
(e) Explain why part (d) needed two pieces of information rather than one.

15.2 Definite Integrals and Area

Set antiderivatives aside for the moment. This section takes up a second and apparently unrelated problem; the two are connected only later.

The problem is area. Regions with straight edges have formulas; a region under a curve has none, because its boundary changes direction at every point. The available approach is approximation.

Cut the interval from a to b into n strips, each of width Δx. Replace each strip by a rectangle whose height is the value of the function somewhere in that strip. Every rectangle has area height $\times$ width, so the total is

$$\sum_{i=1}^{n} f(x_i)\,\Delta x.$$

This is not the area. It is a sum of rectangles, and it differs from the area by whatever those rectangles fail to cover or cover in excess. Narrower strips leave less room for that error, and as n increases the sum settles on a single value.

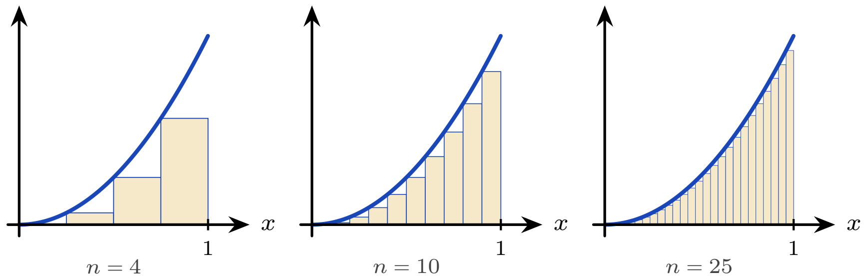

Figure 15.1 Rectangles under $y = x^2$ on $[0, 1]$, taking the height at the left of each strip. Four strips give 0.219, ten give 0.285 and twenty-five give 0.314; the true area is $\frac{1}{3}$. The missing pieces above each rectangle shrink as the strips narrow.

The definition allows the height to be read anywhere in the strip, but two choices are natural: the value at the left end, or the value at the right. Taking both is what makes the argument work, because they err in opposite directions.

For a curve that rises across the interval, as $y = x^2$ does on $[0, 1]$, every left-hand rectangle falls short of the curve and every right-hand one overshoots it. The first sum therefore underestimates the area and the second overestimates it, so the true value is trapped between them. For a curve that falls across the interval the two exchange roles, and the bracketing still holds.

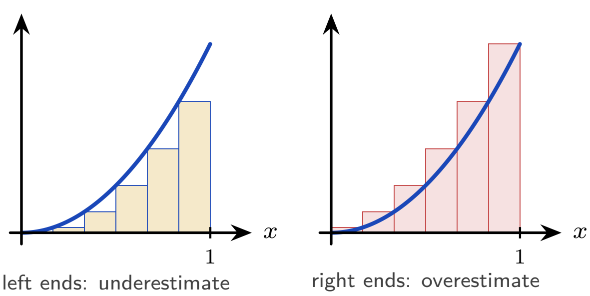

Figure 15.2 The same six strips, with the height of each rectangle read at the left of its strip and then at the right. On a rising curve the left-hand rectangles all sit below it and the right-hand ones all rise above it, so the area lies between the two sums. Narrowing the strips closes the gap from both sides at once.

The numbers behave as the picture predicts.

n	4	10	100	1000
low estimate	0.219	0.285	0.328	0.3328
high estimate	0.469	0.385	0.338	0.3338

The two rows converge, and the value they enclose is $\frac{1}{3}$. That limit is what is meant by the area, and it is called the **definite integral**.

DEF The **definite integral** of f from a to b is the limit of the rectangle sums as the strips become arbitrarily narrow:

$$\int_a^b f(x)\,dx = \lim_{\Delta x \to 0} \sum_i f(x_i)\,\Delta x.$$

For $f(x) \geq 0$ it is the area between the curve and the x-axis from a to b.

The notation can now be read directly. The elongated S denotes a sum, $f(x)\,dx$ represents a single rectangle of height $f(x)$ and vanishing width dx, and the integral accumulates them.

As it stands, however, the definition is of no computational use. Nothing in it indicates how such a limit is to be evaluated, and summing a thousand rectangles by hand is not a practical method.

The **Fundamental Theorem of Calculus** supplies the missing method. It states that this limit of sums can be evaluated by antidifferentiation, so the two ideas of the chapter are in fact one calculation.

KEY **The Fundamental Theorem of Calculus.**

$$\int_a^b f(x)\,dx = \Big[F(x)\Big]_a^b = F(b) - F(a), \qquad \text{where } F' = f.$$

Reversing a derivative and adding up rectangles give the same answer. The constant C cancels in the subtraction, which is why a definite integral is a plain number while an indefinite one carries a $+C$. Why the theorem is true is shown in §15.2.1; the rest of the chapter needs only the statement.

KEY · IB

$$A = \int_a^b y\,dx \qquad (y \geq 0)$$

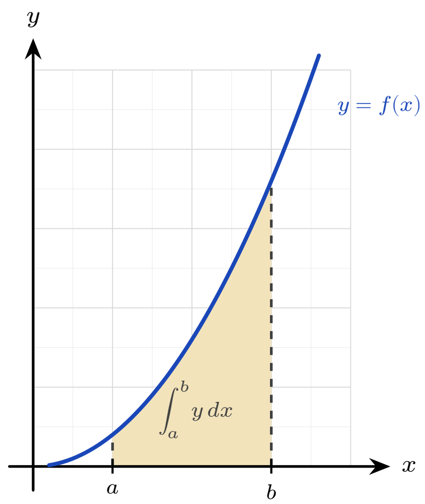

Figure 15.3 For $y \geq 0$ the definite integral measures the shaded area between the curve and the x-axis, from $x = a$ to $x = b$.

EXAMPLE 15.3

Evaluate $\displaystyle\int_0^2 3x^2\,dx$.

$$\int_0^2 3x^2\,dx = \Big[x^3\Big]_0^2 = 8 - 0 = 8.$$

TIP Your GDC evaluates definite integrals directly, and on a calculator paper that is often all a question requires. Even when analytic working is demanded, the GDC value is a free check of your antiderivative.

15.2.1 The Fundamental Theorem of Calculus (Extension)

This section shows why the Fundamental Theorem is true. It is not examined, and nothing later depends on reading it.

The claim to be justified is that an area, defined as a limit of sums, can be found by antidifferentiation. Throughout, f is continuous with $f(x) \geq 0$.

Suppose the area has a formula, in the way that πr^2 is a formula for the area of a circle. Fix a left end a and let the right end move: write $A(x)$ for the area under the curve from a to x.

If such a function exists, the area of *any* band follows at once by subtraction. The area from x_1 to x_2 is the area up to x_2 minus the area up to x_1:

$$\text{area from } x_1 \text{ to } x_2 \;=\; A(x_2) - A(x_1).$$

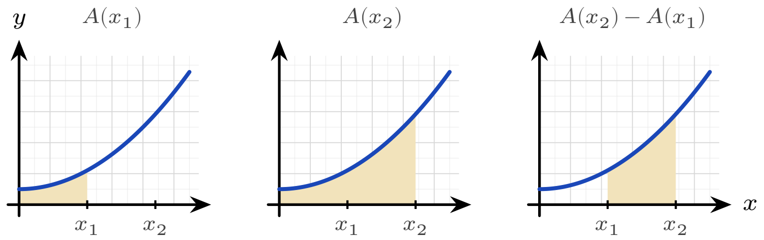

Figure 15.4 If an area function A exists, the area of a band is a difference. Shade up to x_2, remove what was shaded up to x_1, and the band between them is what remains. Finding A would turn every area question into one subtraction.

Nothing is proved yet; this is what would follow *if* A existed. So ask what kind of function A is. When x increases by a small amount h, the extra area $A(x+h) - A(x)$ is one thin strip of width h.

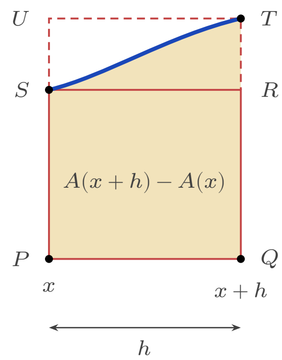

Figure 15.5 One strip, greatly enlarged. Its area is $A(x+h) - A(x)$, and it is trapped: it contains the solid rectangle $PQRS$ and fits inside the dashed rectangle $PQTU$. Both rectangles have width h.

The short rectangle $PQRS$ has the height of the curve at the left of the strip, $f(x)$, and fits inside it. The tall one, $PQTU$, has the height at the right, $f(x+h)$, and contains it. Comparing the three areas,

$$f(x)\,h \;<\; A(x+h) - A(x) \;<\; f(x+h)\,h.$$

Divide through by h:

$$f(x) \;<\; \frac{A(x+h) - A(x)}{h} \;<\; f(x+h).$$

Now let $h \to 0$. The right-hand edge slides back towards the left, so $f(x+h)$ closes on $f(x)$ and the two outer quantities become the same number. The middle is squeezed between them, and it is exactly the definition of $A'(x)$. Therefore

$$A'(x) = f(x).$$

The area function is therefore **an antiderivative of** f — and antiderivatives are something we already know how to find.

The picture is drawn with f rising across the strip. If it falls the two rectangles swap roles, and if it does both, take the smallest and largest values of f on the strip in place of its values at the ends. The squeeze is unchanged.

The computational rule follows at once. If F is any antiderivative of f, then A and F differ by a constant, so $A(x) = F(x) + C$. Setting $x = a$ gives $0 = F(a) + C$, since the area from a to a is nothing, so $C = -F(a)$. Setting $x = b$ then gives

$$\int_a^b f(x)\,dx \;=\; A(b) \;=\; F(b) - F(a),$$

which is the theorem stated in §15.2.

15.2.2 Areas below the axis

Where the curve dips below the x-axis, the integral is negative: it is a signed area. For a true geometric area you integrate the parts separately and take the absolute value of any negative piece.

KEY

$$A = \int_a^b |y|\,dx$$

The booklet writes this second case as $\int y\,dx$ as well, without the modulus. Writing $|y|$ is this book's shorthand for the splitting just described; it is not the printed form, and in an examination you do the splitting yourself.

The distinction is visible as soon as a curve crosses the axis more than once.

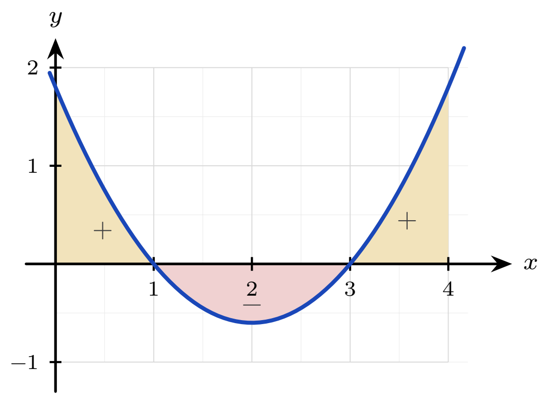

Figure 15.6 A definite integral accumulates *signed* area. The two gold regions lie above the axis and count positively; the scarlet region lies below and counts negatively. A single integral across the whole interval returns the sum of the three signed pieces, which is smaller than the total area enclosed.

EXAMPLE 15.4

Find the area enclosed between $y = x^2 - 4$ and the x-axis from $x = 0$ to $x = 2$.

On $[0, 2]$ the curve lies below the axis, so the integral is negative:

$$\int_0^2 (x^2 - 4)\,dx = \left[\tfrac{x^3}{3} - 4x\right]_0^2 = \tfrac{8}{3} - 8 = -\tfrac{16}{3}.$$

The area is the magnitude, $\dfrac{16}{3}$. An area is never negative, so $-\frac{16}{3}$ answers a different question from the one asked.

CAUTION Before finding an area, check where the curve crosses the x-axis. A curve that goes above and below will have positive and negative parts that partly cancel in a single integral, giving the wrong area. Split at the roots.

YOUR TURN 15.2 Definite Integrals and Area

3. Evaluate each definite integral.

(a) $\displaystyle\int_1^2 4x\,dx$

(b) $\displaystyle\int_0^3 (x^2 + 1)\,dx$

(c) $\displaystyle\int_1^e \frac{1}{x}\,dx$

(d) $\displaystyle\int_1^4 \frac{1}{\sqrt{x}}\,dx$

(e) $\displaystyle\int_0^{\pi/2} \cos x\,dx$

(f) $\displaystyle\int_{-1}^2 (3x^2 - 2x)\,dx$

4.

(a) Find the area under $y = x^2$ between $x = 1$ and $x = 3$.

(b) Find the area enclosed between $y = x^2 - 1$ and the x-axis.

(c) Find the area under $y = \sin x$ from $x = 0$ to $x = \pi$.

(d) Evaluate $\displaystyle\int_0^{2\pi} \sin x\,dx$, and explain why the answer is not the area between the curve and the axis over that interval.

(e) Find the area between $y = x^3$ and the x-axis from $x = -2$ to $x = 2$.

5. The area under $y = x^2$ from $x = 0$ to $x = 2$ is to be estimated by four rectangles of equal width.

(a) Taking the height of each rectangle at the left of its strip, show that the estimate is 1.75.

(b) Taking the height at the right instead, show that the estimate is 3.75.

(c) Find the exact area, and confirm that it lies between the two estimates.

(d) Repeat part (a) with eight rectangles, and show that the left-hand estimate rises to 2.1875.

(e) State what happens to the two estimates as the number of rectangles increases.

15.3 Standard Integrals

Every derivative established in Ch. 13 can be read in reverse. Doing so converts the table of derivatives into a table of integrals, and turns the question "what is the integral of this function?" into the more tractable "what was differentiated to produce it?"

KEY · IB

$$\int \frac{1}{x}\,dx = \ln|x| + C, \qquad \int \sin x\,dx = -\cos x + C$$
$$\int \cos x\,dx = \sin x + C, \qquad \int e^x\,dx = e^x + C$$

None of these is a new fact. Each is a derivative already known, read from right to left, and each should be understood rather than memorised.

- $\int \frac{1}{x}\,dx = \ln|x| + C$ fills the one gap the power rule leaves. Raising $n = -1$ by one gives 0, and the rule then asks you to divide by that new power, which is impossible. So the antiderivative of x^{-1} cannot be a power at all. The logarithm supplies it, because $\frac{d}{dx}\ln x = \frac{1}{x}$. The modulus is there because $\frac{1}{x}$ is defined for negative x as well, and on that side $\frac{d}{dx}\ln(-x) = \frac{-1}{-x} = \frac{1}{x}$ by the chain rule: the same derivative on both sides of the origin.
- $\int e^x\,dx = e^x + C$ because e^x is the function equal to its own derivative. A function that is unchanged by differentiating is unchanged by antidifferentiating.
- $\int \cos x\,dx = \sin x + C$ is the direct reversal of $\frac{d}{dx}\sin x = \cos x$, with no sign to watch.
- $\int \sin x\,dx = -\cos x + C$ carries the minus sign that is most often dropped. Differentiating cosine introduces one: $\frac{d}{dx}\cos x = -\sin x$. To finish with $+\sin x$ you must therefore start from $-\cos x$.

The two trigonometric cases are easiest to keep straight as a cycle. Differentiating moves one step clockwise; integrating moves one step back.

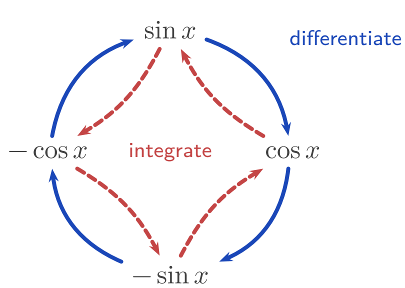

Figure 15.7 The four functions form a closed cycle. Differentiating advances one step clockwise (solid), integrating retreats one step anticlockwise (dashed). Reading the cycle backwards from $\sin x$ lands on

$-\cos x$, which is where the minus sign in $\int \sin x\,dx$ comes from.

At HL, three more standard integrals are provided, the reverses of the exponential and inverse-trigonometric derivatives.

KEY · IB HL

$$\int a^x\,dx = \frac{a^x}{\ln a} + C, \qquad \int \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\arctan\frac{x}{a} + C$$

$$\int \frac{1}{\sqrt{a^2-x^2}}\,dx = \arcsin\frac{x}{a} + C, \qquad |x| < a$$

These three reward the same reading. Differentiating a^x produces an extra factor: writing $a^x = e^{x\ln a}$ and applying the chain rule gives $\frac{d}{dx}a^x = a^x \ln a$. Dividing by $\ln a$ removes it. When $a = e$ the factor $\ln e$ is 1 and the formula collapses to $\int e^x dx = e^x + C$, as it must.

The two inverse-trigonometric forms come from $\frac{d}{dx}\arctan x = \frac{1}{1+x^2}$ and $\frac{d}{dx}\arcsin x = \frac{1}{\sqrt{1-x^2}}$, with a acting as a scale. Their asymmetry is a common source of error: the arctangent form carries a factor $\frac{1}{a}$ and the arcsine form does not. In each case differentiating x/a contributes a chain-rule factor $\frac{1}{a}$; in the arcsine that factor is cancelled by the a released from the square root, and in the arctangent nothing cancels it.

Reading the rest of the HL derivative table backwards gives four more. They must be learned, because the formula booklet gives only the derivatives:

KEY HL

$$\int \sec^2 x\,dx = \tan x + C, \qquad \int \sec x \tan x\,dx = \sec x + C,$$

$$\int \operatorname{cosec}^2 x\,dx = -\cot x + C, \qquad \int \operatorname{cosec} x \cot x\,dx = -\operatorname{cosec} x + C$$

15.3.1 Linear composites

Sometimes the function inside is linear, as in e^{2x} or $\cos 3x$. Integrate as usual, then divide by the coefficient of x inside. This reverses the chain rule, whose extra factor was that same coefficient.

KEY

$$\int f(ax+b)\,dx = \frac{1}{a}F(ax+b) + C, \qquad \text{where } F' = f$$

EXAMPLE 15.5

Find (a) $\int e^{2x}\,dx$, (b) $\int (2x+1)^5\,dx$, (c) $\int \cos 3x\,dx$.
Each has a linear function inside, so integrate and then divide by the coefficient of x:

$$\text{(a) } \tfrac{1}{2}e^{2x} + C, \qquad \text{(b) } \frac{(2x+1)^6}{12} + C, \qquad \text{(c) } \tfrac{1}{3}\sin 3x + C.$$

In (b), reversing the power rule gives $\frac{(2x+1)^6}{6}$, then dividing by the coefficient 2 gives the 12.

EXAMPLE 15.6

Find (a) $\int \frac{1}{x^2+25}\,dx$, (b) $\int \frac{1}{\sqrt{9-x^2}}\,dx$.
Match the standard forms with $a = 5$ and $a = 3$:

$$\text{(a) } \tfrac{1}{5}\arctan\frac{x}{5} + C, \qquad \text{(b) } \arcsin\frac{x}{3} + C.$$

The only work is recognising a^2 as 25 and as 9. After that the standard form gives the answer.

The linear-composite rule extends to the whole table. For instance, $\int \sec^2(2x+5)\,dx = \frac{1}{2}\tan(2x+5) + C$, the guide's own example: integrate as usual, divide by the inner coefficient.

Definite integrals of these standard forms are evaluated exactly as in §15.2: find the antiderivative, then substitute the limits.

EXAMPLE 15.7

Evaluate (a) $\int_0^{\pi/2} \sin x\,dx$, (b) $\int_1^{e^2} \frac{1}{x}\,dx$.
(a) The antiderivative of $\sin x$ is $-\cos x$, so

$$\int_0^{\pi/2} \sin x\,dx = \Big[-\cos x\Big]_0^{\pi/2} = -\cos\frac{\pi}{2} + \cos 0 = 0 + 1 = 1.$$

The area under one quarter-wave of the sine curve is exactly 1, with no π in the answer.
(b) The antiderivative of $\frac{1}{x}$ is $\ln|x|$, and both limits are positive:

$$\int_1^{e^2} \frac{1}{x}\,dx = \Big[\ln x\Big]_1^{e^2} = \ln e^2 - \ln 1 = 2 - 0 = 2.$$

The logarithm hands back the exponent. Stopping at $x = e$ instead would give an area of exactly one, and that is where the number e comes from: it is the upper limit that makes the area under $y = \frac{1}{x}$, starting at $x = 1$, equal to one.

YOUR TURN 15.3 Standard Integrals

6. Find each indefinite integral.

(a) $\int (2\cos x - \sin x)\, dx$

(b) $\int \left(3e^x + \frac{1}{x}\right) dx$

(c) $\int e^{3x}\, dx$

(d) $\int (3x-2)^4\, dx$

(e) $\int \sin 2x\, dx$

(f) $\int \sec^2(4x)\, dx$

(g) $\int \cos(3x+1)\, dx$

(h) $\int \frac{1}{2x+5}\, dx$

7. Find each indefinite integral, matching it to a standard form.

(a) $\int \frac{1}{x^2+9}\, dx$

(b) $\int \frac{1}{\sqrt{16-x^2}}\, dx$

(c) $\int 5^x\, dx$

(d) $\int \frac{1}{x^2+4x+13}\, dx$

(e) $\int \frac{1}{\sqrt{9-4x^2}}\, dx$

(f) $\int \frac{2}{4+x^2}\, dx$

15.4 Rewriting Before Integrating HL

Many integrands do not match a standard form as they stand, but can be rewritten until they do. Four rewritings cover almost every case: divide, complete the square, split into partial fractions, or replace a squared trigonometric function using a double angle. For the first three, deciding between them takes one look at the denominator.

15.4.1 Divide first

When the numerator has degree at least that of the denominator, the fraction is **improper** and no standard form applies. Divide before anything else. For instance $\frac{x^2}{x+1} = x - 1 + \frac{1}{x+1}$, so the integral is $\frac{x^2}{2} - x + \ln|x+1| + C$. Only when the numerator has the lower degree are the next two methods available.

15.4.2 Completing the square HL

An examination question rarely presents a denominator already in the form a^2+x^2. A quadratic such as x^2+2x+5 has that shape, but only after you complete the square.

EXAMPLE 15.8

Find $\displaystyle\int \frac{1}{x^2+2x+5}\,dx$.

Complete the square: $x^2+2x+5=(x+1)^2+4$. This is the arctangent shape with $a=2$ and the shifted variable $x+1$:

$$\int \frac{dx}{(x+1)^2+4} = \frac{1}{2}\arctan\left(\frac{x+1}{2}\right)+C.$$

Any quadratic with no real roots can be written in the form a^2+u^2 this way. The shift from x to $x+1$ does not change the integral, because the derivative of $x+1$ is 1.

15.4.3 Partial fractions HL

When the quadratic *does* factorise, use a different method. Split the fraction into partial fractions, as in Ch. 6. Each piece then has the form $\frac{1}{x+k}$, and integrates to a logarithm.

EXAMPLE 15.9

Find $\displaystyle\int \frac{1}{x^2+3x+2}\,dx$.

Factor and decompose: $\dfrac{1}{(x+1)(x+2)} = \dfrac{1}{x+1} - \dfrac{1}{x+2}$. Then

$$\int\left(\frac{1}{x+1}-\frac{1}{x+2}\right)dx = \ln|x+1| - \ln|x+2| + C = \ln\left|\frac{x+1}{x+2}\right| + C.$$

Factorisable denominator: partial fractions and logarithms. Irreducible denominator: complete the square and arctangent. Choosing between the two is the first decision, and factorising the denominator is what settles it.

15.4.4 Using a double angle

There is no standard integral for $\sin^2 x$ or $\cos^2 x$, and no amount of staring at the table will produce one. Neither is the derivative of anything simple. The way through is to remove the square before integrating, using the double-angle identity from Ch. 8:

$$\cos 2x = 1-2\sin^2 x = 2\cos^2 x - 1.$$

Rearranged, these give the two rewritings that matter here:

KEY

$$\sin^2 x = \frac{1-\cos 2x}{2}, \qquad \cos^2 x = \frac{1+\cos 2x}{2}$$

Each right-hand side is a constant plus a linear composite, and both of those integrate immediately.

EXAMPLE 15.10

Find (a) $\int \sin^2 x\,dx$, (b) $\int_0^{\pi} \sin^2 x\,dx$.

(a) Replace the square first:

$$\int \sin^2 x\,dx = \int \frac{1-\cos 2x}{2}\,dx = \frac{1}{2}\int (1-\cos 2x)\,dx = \frac{x}{2} - \frac{\sin 2x}{4} + C.$$

The $\frac{1}{4}$ comes from the linear composite: integrating $\cos 2x$ gives $\frac{1}{2}\sin 2x$, and the outer $\frac{1}{2}$ halves it again.

(b) Now use the limits:

$$\int_0^{\pi} \sin^2 x\,dx = \left[\frac{x}{2} - \frac{\sin 2x}{4}\right]_0^{\pi} = \left(\frac{\pi}{2} - 0\right) - (0-0) = \frac{\pi}{2}.$$

The answer comes out cleanly, and there is a reason for that. Over a full arch, $\sin^2 x$ averages exactly $\frac{1}{2}$, so the area is half the width π. The oscillating part, $\sin 2x$, contributes nothing over a whole number of half-periods.

The same rewriting handles $\cos^2 x$, and it is the only way to integrate the square of a sine or cosine at this level. Expect it whenever a volume of revolution involves a trigonometric curve, since the volume formula squares the function.

YOUR TURN 15.4 Rewriting Before Integrating HL

8.

(a) Find $\int \frac{1}{x^2-9}\,dx$ using partial fractions.

(b) Find $\int \frac{4x+5}{(x+1)(x+2)}\,dx$.

(c) Explain how you decide, from the denominator alone, whether to use partial fractions or to complete the square.

9. Rewrite each integrand before integrating.

(a) $\int \frac{x^2+3x}{x}\,dx$

(b) $\int \frac{2x+1}{x}\,dx$

(c) $\int \frac{x^3-1}{x^2}\,dx$

(d) $\int \frac{1}{x^2+6x+13}\,dx$

15.5 Integration by Substitution

Substitution reverses the chain rule. Look for a composite function whose inner function also appears, up to a constant multiple, as a factor outside. Put u equal to that inner function, and a hard integral in x becomes an easy one in u.

The method has four steps. Let u be the inner function. Replace dx using $du = \frac{du}{dx}\,dx$. Integrate in u. Then put x back.

EXAMPLE 15.11

Find $\int 2x(x^2+1)^3\,dx$.

Let $u = x^2+1$, so $\frac{du}{dx} = 2x$, giving $du = 2x\,dx$. The integral becomes

$$\int u^3\,du = \frac{u^4}{4} + C = \frac{(x^2+1)^4}{4} + C.$$

The factor $2x$ outside is exactly the $\frac{du}{dx}$ required. Without it the substitution would leave an x behind and fail.

EXAMPLE 15.12

Find $\int \frac{2x}{x^2+1}\,dx$.

With $u = x^2+1$, $du = 2x\,dx$:

$$\int \frac{1}{u}\,du = \ln|u| + C = \ln(x^2+1) + C.$$

An integral of the form $\frac{f'(x)}{f(x)}$ always integrates to $\ln|f(x)|$.

CAUTION For a definite integral by substitution, either convert the limits to u-values, or substitute back to x before applying the original limits. Applying the x-limits to a u-expression gives a wrong answer.

EXAMPLE 15.13

Evaluate $\int_0^{\sqrt{3}} x\sqrt{x^2+1}\,dx$.

Let $u = x^2+1$, so $du = 2x\,dx$. Convert the limits too: $x = 0 \Rightarrow u = 1$ and $x = \sqrt{3} \Rightarrow u = 4$. Then

$$\int_0^{\sqrt{3}} x\sqrt{x^2+1}\,dx = \frac{1}{2}\int_1^4 \sqrt{u}\,du = \frac{1}{2}\cdot\frac{2}{3}\left[u^{3/2}\right]_1^4 = \frac{1}{3}(8-1) = \frac{7}{3}.$$

Once the limits are converted, you never return to x at all; the whole calculation finishes in u.

Substitution is at its most useful on exponential and trigonometric integrands, where no rearrangement of the standard table will help.

EXAMPLE 15.14

Find (a) $\int xe^{-x^2}\,dx$, (b) $\int \sin^4 x\cos x\,dx$.

(a) The inner function is $-x^2$, and its derivative $-2x$ appears outside up to the constant -2. Let $u = -x^2$, so $du = -2x\,dx$ and $x\,dx = -\frac{1}{2}\,du$:

$$\int xe^{-x^2}\,dx = -\frac{1}{2}\int e^u\,du = -\frac{1}{2}e^u + C = -\frac{1}{2}e^{-x^2} + C.$$

On its own, e^{-x^2} cannot be integrated in terms of the functions of this course. The factor x is what makes the integral possible, and losing it makes the problem unsolvable.

(b) Here the inner function is $\sin x$, and its derivative $\cos x$ is sitting outside. Let $u = \sin x$, so $du = \cos x\,dx$:

$$\int \sin^4 x\cos x\,dx = \int u^4\,du = \frac{u^5}{5} + C = \frac{\sin^5 x}{5} + C.$$

In both parts the test is the same: is the derivative of the inner function present as a factor? If it is, substitution works. If it is not, look for another method.

YOUR TURN 15.5 Integration by Substitution

10. Find each integral by substitution.

(a) $\int 3x^2(x^3+1)^4\,dx$

(b) $\int xe^{x^2}\,dx$

(c) $\int 2x\sqrt{x^2+1}\,dx$

(d) $\int \frac{x}{(x^2+4)^2}\,dx$

(e) $\int \frac{\ln x}{x}\,dx$

(f) $\int \sin^3 x\cos x\,dx$

11.

(a) Evaluate $\int_0^1 3x^2(x^3+1)^4\,dx$ by converting the limits.

(b) Evaluate the same integral by substituting back to x first, and confirm the answers agree.

(c) Evaluate $\int_0^{\pi/2} \sin^2 x\cos x\,dx$ by converting the limits.

15.6 Integration by Parts HL

Substitution reverses the chain rule; **integration by parts** reverses the product rule. It handles products that substitution cannot, such as xe^x or $\ln x$.

KEY · IB HL

$$\int u\,\frac{dv}{dx}\,dx = uv - \int v\,\frac{du}{dx}\,dx$$

You split the integrand into a part u to differentiate and a part $\dfrac{dv}{dx}$ to integrate. In practice it is easier to work with the same rule written in terms of the two functions you can see in front of you. Writing the integrand as a product uv, and letting $\displaystyle\int v\,dx$ stand for any antiderivative of v:

KEY

$$\int uv\,dx \;=\; u\int v\,dx \;-\; \int\left(\int v\,dx\right)\frac{du}{dx}\,dx$$

Read from left to right this says: integrate the second function, multiply by the first, then subtract the integral of that product with the first function differentiated. Nothing new has happened, but v is never named separately.

Everything now depends on which factor you call u. Choose the one that becomes simpler when differentiated, so that the second integral is easier than the one you started with. The order **ILATE** is a reliable guide: whichever type appears first in this list should be u.

- **I** **Inverse trigonometric** — $\arcsin x$, $\arctan x$
- **L** **Logarithmic** — $\ln x$
- **A** **Algebraic** — x, x^2, $3x - 1$
- **T** **Trigonometric** — $\sin x$, $\cos x$
- **E** **Exponential** — e^x, 2^x

In $\int xe^x\,dx$ the factors are algebraic and exponential, so A comes before E and $u = x$. In $\int x\ln x\,dx$ they are logarithmic and algebraic, so L comes before A and $u = \ln x$. The order works because differentiating reduces the first three types to something simpler, while trigonometric and exponential functions repeat themselves however often you differentiate them.

CAUTION ILATE is a guide, not a theorem. It gives the right choice for the integrals you will meet in this course, but the test that always applies is the one behind it: does the second integral end up easier than the first?

EXAMPLE 15.15

Find $\displaystyle\int xe^x\,dx$.

Let $u = x$ (simplifies to 1) and $\dfrac{dv}{dx} = e^x$ (so $v = e^x$):

$$\int xe^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x + C = e^x(x-1) + C.$$

Choosing $u = x$ is what makes the new integral, $\int e^x\,dx$, trivial. The other choice would make it worse.

EXAMPLE 15.16

Find (a) $\displaystyle\int (x+1)\sin x\,dx$, (b) $\displaystyle\int_0^{\pi/2} (x+1)\cos x\,dx$.

(a) The factors are algebraic and trigonometric, so ILATE gives $u = x{+}1$ and $\dfrac{dv}{dx} = \sin x$, hence $v = -\cos x$:

$$\int (x+1)\sin x\,dx = -(x+1)\cos x - \int(-\cos x)\,dx = -(x+1)\cos x + \sin x + C.$$

Watch the two minus signs. Integrating $\sin x$ gives $-\cos x$, and the formula then subtracts the integral of that, which turns the sign back.

(b) Same choice, with $v = \sin x$:

$$\int (x+1)\cos x\,dx = (x+1)\sin x - \int \sin x\,dx = (x+1)\sin x + \cos x + C.$$

Now apply the limits:

$$\int_0^{\pi/2} (x+1)\cos x\,dx = \Big[(x+1)\sin x + \cos x\Big]_0^{\pi/2} = \left(\frac{\pi}{2} + 1 + 0\right) - (0+1) = \frac{\pi}{2}.$$

For a definite integral by parts, find the whole antiderivative first, then substitute the limits once at the end.

EXAMPLE 15.17

Find $\displaystyle\int \ln x\,dx$.

There is only one function, but write $u = \ln x$ and $\dfrac{dv}{dx} = 1$ (so $v = x$):

$$\int \ln x\,dx = x\ln x - \int x\cdot\frac{1}{x}\,dx = x\ln x - x + C.$$

Taking $\dfrac{dv}{dx} = 1$ is a standard move. It is how both $\ln x$ and $\arctan x$ are integrated.

15.6.1 Repeated integration by parts

If one application of parts leaves an integral that still needs parts, apply it again. Each round of differentiation reduces the power of x by one.

EXAMPLE 15.18

Find $\int x^2 e^{-x}\,dx$.

First round, $u = x^2$ and $\frac{dv}{dx} = e^{-x}$, so $v = -e^{-x}$:

$$\int x^2 e^{-x}\,dx = -x^2 e^{-x} + 2\int x e^{-x}\,dx.$$

The remaining integral needs parts a second time, with $u = x$:

$$\int x e^{-x}\,dx = -x e^{-x} + \int e^{-x}\,dx = -e^{-x}(x+1) + C.$$

Substituting that back,

$$\int x^2 e^{-x}\,dx = -x^2 e^{-x} - 2e^{-x}(x+1) + C = -e^{-x}\left(x^2 + 2x + 2\right) + C.$$

A power x^n takes n rounds of parts to clear, since only the power changes from round to round; the exponential returns unchanged each time.

Something cleverer happens when *neither* factor ever simplifies. For $e^x \sin x$, two rounds of parts bring the original integral back, and you solve for it like an equation.

EXAMPLE 15.19

Find $I = \int e^x \sin x\,dx$.

Parts with $u = \sin x$:

$$I = e^x \sin x - \int e^x \cos x\,dx.$$

Parts again on the new integral, $u = \cos x$:

$$\int e^x \cos x\,dx = e^x \cos x + \int e^x \sin x\,dx = e^x \cos x + I.$$

Substitute back:

$$I = e^x \sin x - e^x \cos x - I \implies 2I = e^x(\sin x - \cos x) \implies I = \tfrac{1}{2}e^x(\sin x - \cos x) + C.$$

The original integral reappeared on the right-hand side, which turned the problem into an equation for I. Watch for this whenever neither factor gets simpler: two rounds of parts bring I back, and algebra finishes the job.

15.6.2 Choosing a method

Five methods are now available, and most marks lost in this topic are lost by starting the wrong one. The choice is settled by looking at the integrand before writing anything.

What you see	Method	Example
A standard form exactly, or with a linear inside	read it off, dividing by the inner coefficient	$\cos 3x$, e^{2x}, $(2x+1)^5$
A function together with a multiple of its own derivative	substitution	$2x(x^2+1)^3$, $\dfrac{2x}{x^2+1}$
A product of two unlike functions, one of which simplifies when differentiated	parts	xe^x, $x\sin x$, $\ln x$
A fraction whose denominator factorises	partial fractions, then logarithms	$\dfrac{1}{x^2+3x+2}$
A fraction whose denominator is a quadratic with no real roots	complete the square, then arctangent	$\dfrac{1}{x^2+2x+5}$

Work down the list in that order, because the earlier tests are quicker to apply and the earlier methods are shorter. Two habits help further. Check first whether the fraction is improper, since dividing may remove the problem entirely. And if a substitution leaves an integral no easier than the one you started with, the substitution was the wrong one; go back rather than pressing on.

YOUR TURN 15.6 Integration by Parts HL

12. Find each integral by parts.

(a) $\displaystyle\int x\cos x\,dx$

(b) $\displaystyle\int xe^{2x}\,dx$

(c) $\displaystyle\int x\ln x\,dx$

(d) $\displaystyle\int \arctan x\,dx$

(e) $\displaystyle\int x^2\ln x\,dx$

(f) $\displaystyle\int x\sec^2 x\,dx$

13.

(a) Evaluate $\displaystyle\int_0^{\pi} x \sin x \, dx$.

(b) Evaluate $\displaystyle\int_0^{1} x e^x \, dx$.

(c) State which factor you chose as u in part (b), and why the other choice would not help.

(d) Use ILATE to say which factor should be u in each of $\displaystyle\int x^2 e^x \, dx$, $\displaystyle\int x \arctan x \, dx$ and $\displaystyle\int e^x \sin x \, dx$.

(e) Explain why the third integral in part (d) behaves differently from the first two.

15.7 Areas Between Curves

To find the area between two curves, integrate the difference: the upper curve minus the lower. The signed-area problem disappears, because the difference is positive wherever the first curve is on top, regardless of the axis.

KEY

$$A = \int_a^b \left(y_{\text{top}} - y_{\text{bottom}}\right) dx$$

The limits a and b are usually the x-coordinates where the curves intersect, found by setting them equal.

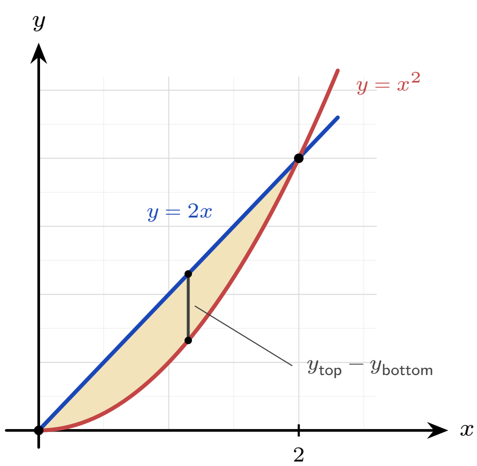

Figure 15.8 The region enclosed between $y = 2x$ and $y = x^2$. The curves meet where $2x = x^2$, at $x = 0$ and $x = 2$, and the line is the upper boundary in between, so the area is $\int_0^2 (2x - x^2)\, dx$.

EXAMPLE 15.20

Find the area enclosed between $y = 2x$ and $y = x^2$.

Intersections: $2x = x^2 \Rightarrow x = 0$ or $x = 2$. On $(0, 2)$ the line $y = 2x$ is above, so

$$A = \int_0^2 (2x - x^2)\, dx = \left[x^2 - \tfrac{x^3}{3}\right]_0^2 = 4 - \tfrac{8}{3} = \tfrac{4}{3}.$$

15.7.1 Area between a curve and the y-axis HL

Sometimes it is easier to integrate with respect to y: express x as a function of y and accumulate horizontal slices. This gives the area between the curve and the y-axis.

KEY · IB HL

$$A = \int_a^b x\, dy$$

EXAMPLE 15.21

Find the area between the curve $x = y^2$ and the y-axis from $y = 0$ to $y = 2$.

$$A = \int_0^2 y^2\, dy = \left[\tfrac{y^3}{3}\right]_0^2 = \tfrac{8}{3}.$$

Integrating in y avoids splitting the region, which would be awkward in x.

Nothing in the method requires the curves to be polynomials. The steps are the same for trigonometric and exponential boundaries: find where they cross, decide which is on top, and integrate the difference.

EXAMPLE 15.22

Find the area enclosed between $y = \sin x$ and $y = \cos x$ between their first two intersections for $x \geq 0$.

They meet where $\sin x = \cos x$, that is $\tan x = 1$, so the first two intersections are at $x = \frac{\pi}{4}$ and $x = \frac{5\pi}{4}$.

On $\left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$ the sine is above: at $x = \frac{\pi}{2}$, $\sin\frac{\pi}{2} = 1$ while $\cos\frac{\pi}{2} = 0$. So

$$A = \int_{\pi/4}^{5\pi/4} (\sin x - \cos x)\, dx = \Big[-\cos x - \sin x\Big]_{\pi/4}^{5\pi/4}.$$

At the upper limit both $\cos x$ and $\sin x$ equal $-\frac{\sqrt{2}}{2}$, so the bracket is $\sqrt{2}$; at the lower limit both equal $\frac{\sqrt{2}}{2}$ and the bracket is $-\sqrt{2}$:

$$A = \sqrt{2} - \left(-\sqrt{2}\right) = 2\sqrt{2} \approx 2.83.$$

Deciding which curve is on top is the step that carries the sign. Testing one convenient point inside the interval, here $x = \frac{\pi}{2}$, settles it.

YOUR TURN 15.7 Areas Between Curves

14. Find each enclosed area.

(a) Between $y = x$ and $y = x^2$.
(b) Between $y = 4 - x^2$ and the x-axis.
(c) Between $y = 6 - x^2$ and $y = 2$.
(d) Between $x = y^2$ and the y-axis, from $y = 0$ to $y = 3$.
(e) Between $y = x^2$ and $y = 2 - x^2$.

15. Consider the region enclosed by $y = x^2$ and $y = x + 2$.

(a) Find the x-coordinates where the two curves meet.
(b) State which curve is the upper boundary between those values.
(c) Find the area of the enclosed region.
(d) State what answer you would get by integrating $x^2-(x+2)$ instead, and explain what has gone wrong.

16. Find each enclosed area.

(a) Under $y = \cos x$, from $x = 0$ to $x = \frac{\pi}{2}$.
(b) Under $y = e^x$, from $x = 0$ to $x = 2$.
(c) Between $y = \sin x$ and the x-axis, from $x = 0$ to $x = \pi$.
(d) Between $y = e^x$ and $y = e^{-x}$, from $x = 0$ to $x = 1$.
(e) Under $y = \dfrac{1}{x}$, from $x = 1$ to $x = 4$.

15.8 Volumes of Revolution HL

Rotate a curve about an axis and it sweeps out a solid. To find the volume, slice the solid into thin disks perpendicular to the axis. Each disk is a cylinder of radius y (the curve's height) and thickness dx, with volume $\pi y^2\,dx$. Summing the disks and letting their thickness shrink turns the sum into an integral.

KEY · IB HL

$$V = \int_a^b \pi y^2\,dx \quad \text{(about the } x\text{-axis)}, \qquad V = \int_a^b \pi x^2\,dy \quad \text{(about the } y\text{-axis)}$$

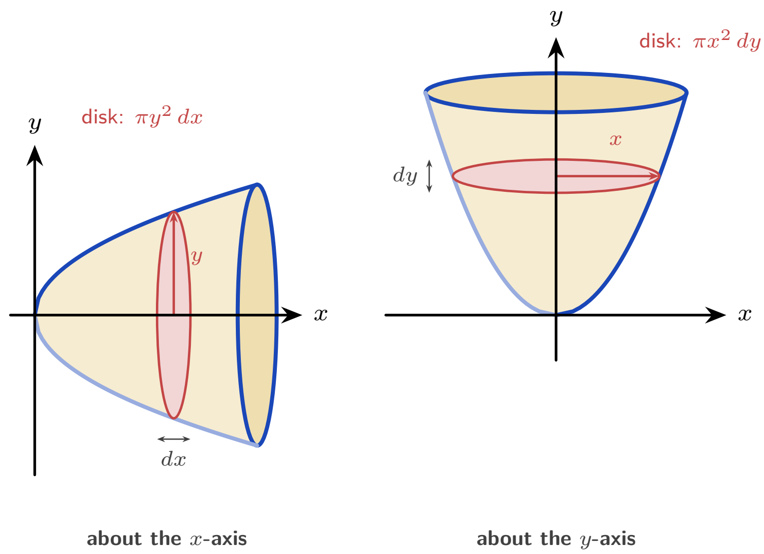

Figure 15.9 The same curve rotated about each axis. Turning it about the x-axis gives disks stacked along x: each has radius y and thickness dx, so its volume is $\pi y^2\,dx$. Turning it about the y-axis gives disks stacked along y: each has radius x and thickness dy, so its volume is $\pi x^2\,dy$. In both cases the integral adds the disks from one end of the solid to the other.

EXAMPLE 15.23

The curve $y = x^3$ for $0 \le x \le 2$ is rotated about the x-axis. Find the volume.

$$V = \int_0^2 \pi (x^3)^2\,dx = \pi \int_0^2 x^6\,dx = \pi \left[\tfrac{x^7}{7}\right]_0^2 = \frac{128\pi}{7}.$$

Square the function *before* integrating; squaring after is wrong.

EXAMPLE 15.24

The region between $y = x^2$ and the y-axis, from $y = 0$ to $y = 1$, is rotated about the y-axis. Find the volume.

About the y-axis, slice into disks of radius x and thickness dy, with $x^2 = y$:

$$V = \int_0^1 \pi x^2\,dy = \pi \int_0^1 y\,dy = \pi \left[\tfrac{y^2}{2}\right]_0^1 = \frac{\pi}{2}.$$

The limits are now y-values, and x^2 is rewritten in terms of y.

Read the region carefully. A disk of radius x is swept by the strip running from the y-*axis* out to the curve, so the region rotated is the one between the curve and the y-axis, not the one under the curve. Those are two different regions and in general they give two different volumes.

Because the formula squares the function, a trigonometric curve produces $\sin^2$ or $\cos^2$, and those must be rewritten with a double angle before they can be integrated See §15.4.

EXAMPLE 15.25

The region under $y = \sin 2x$ from $x = 0$ to $x = \frac{\pi}{4}$ is rotated 360° about the x-axis. Find the volume.

Apply the formula, squaring the function first:

$$V = \int_0^{\pi/4} \pi y^2 \, dx = \pi \int_0^{\pi/4} \sin^2 2x \, dx.$$

There is no standard integral for $\sin^2 2x$, so replace it using $\sin^2 2x = \dfrac{1 - \cos 4x}{2}$:

$$\pi \int_0^{\pi/4} \frac{1 - \cos 4x}{2} \, dx = \pi \left[\frac{x}{2} - \frac{\sin 4x}{8} \right]_0^{\pi/4} = \pi \left(\frac{\pi}{8} - 0 \right) = \frac{\pi^2}{8} \approx 1.23.$$

Two πs appear for different reasons, and they must be kept apart: one comes from the volume formula, the other from the upper limit.

EXAMPLE 15.26

The region under $y = e^x$ from $x = 0$ to $x = 2$ is rotated about the x-axis. Find the volume.

$$V = \pi \int_0^2 (e^x)^2 \, dx = \pi \int_0^2 e^{2x} \, dx = \pi \left[\tfrac{1}{2} e^{2x} \right]_0^2 = \frac{\pi}{2} (e^4 - 1) \approx 84.2.$$

Squaring first is what turns e^x into e^{2x}, a linear composite. Squaring the integral instead of the function is the usual error, and it gives a different answer entirely.

YOUR TURN 15.8 Volumes of Revolution HL

17. Find each volume of revolution about the x-axis.

(a) $y = x^2$, $0 \le x \le 2$.
(b) $y = \sqrt{x}$, $0 \le x \le 4$.
(c) $y = 2x$, $0 \le x \le 3$.
(d) $y = \dfrac{1}{x}$, $1 \le x \le 3$.
(e) $y = \sin x$, $0 \le x \le \pi$.

18.

(a) The region between $y = x^2$ and the y-axis, from $y = 0$ to $y = 4$, is rotated about the y-axis. Find the volume.

(b) Explain why the answer differs from the earlier volume about the x-axis, even though the same curve is used.

(c) The region between $x = y^2$ and the y-axis, from $y = 0$ to $y = 2$, is rotated about the y-axis. Find the volume.

19. Each of these volumes of revolution about the x-axis needs a rewriting before it can be integrated.

(a) $y = \cos x$, $0 \le x \le \frac{\pi}{2}$.

(b) $y = e^x$, $0 \le x \le 1$.

(c) $y = e^{-x}$, $0 \le x \le \ln 2$.

(d) $y = \sin x$, $0 \le x \le \frac{\pi}{2}$.

(e) Explain why a trigonometric curve always forces the double-angle rewriting of §15.4, while an exponential one does not.

15.9 Kinematics by Integration

In Ch. 14 you differentiated displacement to get velocity, and velocity to get acceleration. Integration reverses those two steps. Integrate acceleration to get velocity, and velocity to get displacement, using an initial condition each time to fix the constant.

KEY · IB

$$\text{displacement} = \int_{t_1}^{t_2} v\,dt, \qquad \text{distance travelled} = \int_{t_1}^{t_2} |v|\,dt$$

Displacement is the signed change in position; distance is the total path length. They differ whenever the particle reverses direction, because backward motion subtracts from displacement but adds to distance. On a velocity–time graph the difference is visible: area below the axis counts as negative for displacement, and as positive for distance.

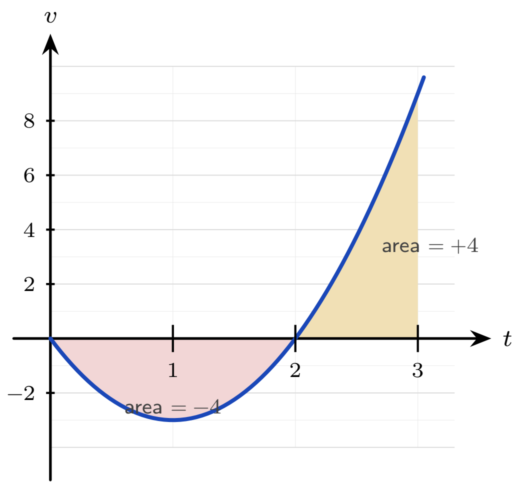

Figure 15.10 The velocity $v = 3t^2 - 6t$ over $0 \le t \le 3$. The particle moves backwards until $t = 2$ and forwards after it. Displacement adds the two signed areas, $-4 + 4 = 0$; distance adds their sizes, $4 + 4 = 8$ m.

EXAMPLE 15.27

A particle has velocity $v(t) = 3t^2 - 6t\,\mathrm{m\,s^{-1}}$. Find its displacement and the distance travelled over $0 \le t \le 3$.

Displacement integrates v directly:

$$\int_0^3 (3t^2 - 6t)\,dt = \Big[t^3 - 3t^2\Big]_0^3 = 27 - 27 = 0.$$

The particle ends where it began. For distance, note $v = 3t(t-2)$ is negative on $(0, 2)$ and positive on $(2, 3)$, so split:

$$\left|\int_0^2 v\,dt\right| + \int_2^3 v\,dt = |-4| + 4 = 8 \text{ m}.$$

Zero displacement but 8 m travelled: the particle went out 4 m and came back.

Oscillating motion is handled the same way, and it makes the difference between displacement and distance especially clear.

EXAMPLE 15.28

A particle has velocity $v(t) = 4\cos 2t\,\mathrm{m\,s^{-1}}$. Find its displacement and the distance travelled over $0 \le t \le \frac{\pi}{2}$ seconds.

Displacement integrates v directly. The antiderivative of $4\cos 2t$ is $2\sin 2t$:

$$\int_0^{\pi/2} 4\cos 2t\,dt = \Big[2\sin 2t\Big]_0^{\pi/2} = 2\sin\pi - 2\sin 0 = 0.$$

The particle finishes where it started. For the distance, first find where it turns: $v = 0$ when $\cos 2t =$

0, so $2t = \frac{\pi}{2}$ and $t = \frac{\pi}{4}$. Split the interval there:

$$\int_0^{\pi/4} 4\cos 2t\,dt = \Big[2\sin 2t\Big]_0^{\pi/4} = 2, \qquad \int_{\pi/4}^{\pi/2} 4\cos 2t\,dt = 0 - 2 = -2.$$

Distance adds the sizes: $|2| + |-2| = 4$ m. So the particle travels 2 m out and 2 m back, ending at its starting point.

The turning point is what makes the split necessary, and forgetting it is the standard error: integrating $|v|$ straight through as if it were v would return 0 for the distance as well.

YOUR TURN 15.9 Kinematics by Integration

20. A particle moves in a straight line.

(a) Its velocity is $v = 2t + 1$. Find its displacement over $0 \le t \le 3$.
(b) Its acceleration is $a = 6t$ and $v(0) = 1$. Find $v(t)$.
(c) Its velocity is $v = t^2 - 4$. Find its displacement over $0 \le t \le 3$.
(d) Its velocity is $v = 4 - 2t$. Find the distance travelled over $0 \le t \le 4$.
(e) For the particle in part (d), find the displacement over the same interval, and explain why the two answers differ.
(f) Its velocity is $v = 3t^2 - 12$. Find the time at which it changes direction, and the distance travelled over $0 \le t \le 3$.

21. A particle moves in a straight line.

(a) Its velocity is $v = 3\cos t\ \mathrm{m\,s^{-1}}$. Find its displacement over $0 \le t \le \frac{\pi}{2}$.
(b) Its velocity is $v = 6e^{-2t}\ \mathrm{m\,s^{-1}}$. Find its displacement over $0 \le t \le 1$.
(c) Its acceleration is $a = \sin t\ \mathrm{m\,s^{-2}}$ and $v(0) = 0$. Find $v(t)$.
(d) Its velocity is $v = 2\cos t\ \mathrm{m\,s^{-1}}$. Find the time in $0 \le t \le \pi$ at which it changes direction, and the distance travelled over that interval.

Chapter 15 · Integral Calculus

Reflections

TOK The proof in §15.2 establishes *that* the area under a curve and the reverse of a derivative are the same calculation. It does not explain *why* the world should be arranged that way. Two ideas invented for different purposes, one geometric and one algebraic, turned out to be one idea.

Does that make the connection a discovery about quantities and their rates, or a consequence of how we chose to define both? Is mathematics invented or uncovered?

IM Long before calculus, Archimedes (3rd century BCE, Syracuse) found the area of a parabolic segment by his "method of exhaustion": filling the region with ever-smaller triangles and summing them. He effectively computed an integral two thousand years before Newton and Leibniz gave the operation a name and a reverse-differentiation shortcut. The idea of accumulation by slicing is older than the algebra we use to carry it out, and his triangles are the ancestors of the rectangles in §15.2.

RW Integration is summation made continuous, so it appears wherever a total is built from a varying rate. The total charge from a varying current, the work done by a changing force, the area of an irregular plot of land, the probability accumulated under a distribution curve (Ch. 12): each is an integral. Whenever a quantity is the running total of a rate, integration computes it.

RW A shape made by spinning a profile about an axis, such as a bottle, a lens or a turbine housing, is fixed completely by that one curve, so its volume is a single integral rather than a measurement. Engineers size fuel tanks and machined parts this way, and the same integral gives the mass once the density is known.

EXT Not every function has an antiderivative in terms of familiar functions: $\int e^{-x^2}\,dx$, central to statistics, cannot be written with powers, exponentials, and trigonometric functions. Yet the definite integral still exists as a perfectly real area, and is computed numerically. The reach of "find the area" is wider than the reach of "reverse the derivative."

End-of-Chapter Problems — Chapter 15: Integral Calculus

1. A curve has $\frac{dy}{dx} = 3x^2 - 4x + 1$ and passes through $(1, 2)$.

(a) Find the equation of the curve. [3]

(b) Find the value of y when $x = 2$. [2]

2. Consider $y = x^2 - 4$.

(a) Find the x-intercepts. [2]

(b) Find the area enclosed between the curve and the x-axis. [4]

3. The curves $y = x^2$ and $y = 8 - x^2$ intersect at two points.

(a) Find the coordinates of the intersection points. [3]

(b) Find the area enclosed between the curves. [4]

4.

(a) Using the substitution $u = x^2 + 4$, find $\displaystyle\int \frac{x}{x^2+4}\,dx$. [3]

(b) Hence evaluate $\displaystyle\int_0^2 \frac{x}{x^2+4}\,dx$, giving an exact answer. [3]

5.

(a) Find $\displaystyle\int xe^{2x}\,dx$ using integration by parts. [3]

(b) Evaluate $\displaystyle\int_0^1 xe^{2x}\,dx$, giving an exact answer. [2]

6. The region bounded by $y = \sqrt{x}$, the x-axis, and $x = 4$ is rotated 360° about the x-axis.

(a) Write down an integral for the volume. [2]

(b) Find the volume. [2]

7. A particle starts at the origin with velocity $5\,\mathrm{m\,s^{-1}}$ and has acceleration $a = 6t - 4\,\mathrm{m\,s^{-2}}$.

(a) Find the velocity $v(t)$. [2]

(b) Find the displacement $s(t)$. [2]

(c) Find the displacement when $t = 2$. [2]

8. Consider the region enclosed by $y = x^2$, the y-axis, and $y = 4$.

(a) Find the area of the region. [3]

(b) Find the volume when this region is rotated about the y-axis. [3]

9.

(a) Find $\displaystyle\int \frac{1}{x^2+4}\,dx$. [2]

(b) Evaluate $\displaystyle\int_0^2 \frac{1}{x^2+4}\,dx$, giving an exact answer. [3]

10. The velocity of a particle is $v(t) = 3t^2 - 12t + 9\,\mathrm{m\,s^{-1}}$, $t \geq 0$.

(a) Find the times when the particle is at rest. [2]

(b) Find the total distance travelled in the first 3 seconds. [5]

11. A curve has gradient function $\dfrac{dy}{dx} = 4x - 3$ and passes through the point $(2, 5)$.

(a) Find the equation of the curve. [3]

(b) Write down its y-intercept. [1]

12. Find the following integrals.

(a) $\displaystyle\int e^{3x}\,dx$ [2]

(b) $\displaystyle\int (4x-1)^6\,dx$ [2]

(c) $\displaystyle\int \sin 2x\,dx$ [2]

13. Evaluate $\displaystyle\int_1^4 \left(\frac{2}{\sqrt{x}} + 1\right) dx$. [4]

14. Consider the curve $y = x^2 - 1$ on the interval $0 \leq x \leq 2$.

(a) Find where the curve crosses the x-axis in this interval. [1]

(b) Find the total area enclosed between the curve and the x-axis on the interval. [5]

15. Find the area of the region enclosed by the curve $y = 4x - x^2$ and the line $y = x$. [5]

16. Using the substitution $u = x^2 + 1$, evaluate $\displaystyle\int_0^2 x(x^2+1)^3\,dx$. [5]

17. Evaluate $\displaystyle\int_0^2 \frac{x}{x^2+4}\,dx$, giving your answer in the form $a \ln b$. [5]

18. Find $\displaystyle\int \frac{x+7}{(x-1)(x+3)}\,dx$. [6]

19. Find $\displaystyle\int \frac{1}{x^2-4x+13}\,dx$. [5]

20. Find the following integrals.

(a) $\displaystyle\int x\cos x\,dx$ [3]

(b) $\displaystyle\int x^2\ln x\,dx$ [4]

21. The region under $y=\sqrt{x}$ from $x=0$ to $x=4$ is rotated about the x-axis.

(a) Find the volume generated. [3]

(b) The same curve, written $x=y^2$ for $0\le y\le 2$, is rotated about the y-axis. Find the volume generated. [4]

22. A particle moves with velocity $v(t)=t^2-4t+3\,\mathrm{m\,s^{-1}}$ for $0\le t\le 4$.

(a) Find the displacement of the particle over the interval. [3]

(b) Find the total distance travelled. [5]

23. The area under $y=x^2+1$ from $x=0$ to $x=3$ is to be estimated by three rectangles of width 1.

(a) Find the estimate obtained by taking the height of each rectangle at the left of its strip. [3]

(b) Find the estimate obtained by taking the height at the right. [2]

(c) Evaluate $\displaystyle\int_0^3 (x^2+1)\,dx$ and verify that it lies between the two estimates. [3]

(d) State, with a reason, whether the left estimate would increase or decrease if six rectangles were used instead. [2]

24. The curve $y=x^2$ and the line $y=2-x$ enclose a region R.

(a) Find the x-coordinates of the two points of intersection. [3]

(b) Sketch the curve and the line, and shade R. [2]

(c) Show that the area of R is $\dfrac{9}{2}$. [5]

(d) Explain why integrating $y_{\text{curve}}-y_{\text{line}}$ instead would give the wrong sign. [2]

25. This question uses integration by parts throughout.

(a) Find $\displaystyle\int x\ln x\,dx$. [4]

(b) Hence evaluate $\int_1^e x \ln x\, dx$, giving your answer in terms of e. [3]

(c) Evaluate $\int_0^{\pi/2} x \cos x\, dx$. [5]

(d) In part (c), state which factor you took as u, and explain what would go wrong with the other choice. [2]

26. Consider the curve $y = x^3$ for $0 \leq x \leq 1$.

(a) The region between the curve and the x-axis is rotated 360° about the x-axis. Show that the volume is $\frac{\pi}{7}$. [4]

(b) The region between the curve and the y-axis is rotated 360° about the y-axis. Find this volume, giving an exact answer. [5]

(c) State which of the two solids has the greater volume, and explain the answer by describing the two regions. [3]

27. A particle moves in a straight line with acceleration $a = 6 - 2t\,\mathrm{m\,s^{-2}}$, and is at rest at $t = 0$.

(a) Show that $v = 6t - t^2$. [3]

(b) Find the time at which the particle is next at rest. [2]

(c) Find the maximum velocity, and the time at which it occurs. [3]

(d) Find the distance travelled from $t = 0$ until the particle is next at rest. [4]

(e) Explain why, for this motion, the distance travelled equals the displacement. [2]

28. Each of these integrals needs a different method. In each part, name the method before using it.

(a) $\int \frac{x^2+1}{x+1}\, dx$ [5]

(b) $\int_1^2 \frac{2x-1}{x^2-x+1}\, dx$ [4]

(c) $\int_0^{\pi/2} \sin x \cos^2 x\, dx$ [4]

(d) $\int \frac{1}{x^2-4x+13}\, dx$ [4]

29. The region R is enclosed by $y = \sin x$, $y = \cos x$ and the y-axis.

(a) Show that the two curves first meet at $x = \frac{\pi}{4}$. [2]

(b) Find the area of R. [4]

(c) R is rotated 360° about the x-axis. Show that the volume is $\frac{\pi}{2}$. [6]

30. The region under $y = \cos x$ from $x = 0$ to $x = \frac{\pi}{2}$ is rotated 360° about the x-axis.

(a) Write down an integral for the volume. [2]

(b) Use a double angle to show that the volume is $\dfrac{\pi^2}{4}$. [5]

(c) The same region under $y = \sin x$ gives the same volume. Explain why, without integrating. [2]

31. A particle moves in a straight line with velocity $v(t) = 6\sin 2t\,\mathrm{m\,s^{-1}}$, for $0 \le t \le \pi$ seconds.

(a) Find the times at which the particle is at rest. [3]

(b) Find the displacement over $0 \le t \le \pi$. [3]

(c) Find the total distance travelled over the same interval. [4]

(d) Explain why the two answers differ. [2]

32. A particle starts from rest at the origin with acceleration $a(t) = 4e^{-2t}\,\mathrm{m\,s^{-2}}$.

(a) Show that $v(t) = 2 - 2e^{-2t}$. [3]

(b) Find the displacement over $0 \le t \le 1$. [4]

(c) State the velocity the particle approaches as t grows large, and explain what is happening physically. [2]

33. Consider the region enclosed by $y = e^x$, $y = e^{-x}$ and the line $x = 1$.

(a) Show that the two curves meet at $x = 0$. [1]

(b) Find the area of the region. [4]

(c) The region is rotated 360° about the x-axis. Find the volume, giving your answer in terms of e and π. [5]

34. Each of these integrals needs a different method. In each part, name the method before using it.

(a) $\displaystyle\int xe^{3x}\,dx$ [4]

(b) $\displaystyle\int \cos^2 3x\,dx$ [4]

(c) $\displaystyle\int \frac{\sin x}{\cos x}\,dx$ [3]

(d) $\displaystyle\int_0^{\ln 2} \frac{e^x}{1+e^x}\,dx$ [4]

Challenge Questions

35. Solids you already know. Every volume formula from Ch. 7 can be recovered by rotating a curve. This question derives three of them.

(a) The line $y = r$ for $0 \le x \le h$ is rotated about the x-axis. Show that the volume is $\pi r^2 h$, and name the solid. [3]

(b) The line $y = \dfrac{r}{h}x$ for $0 \le x \le h$ is rotated about the x-axis. Show that the volume is $\frac{1}{3}\pi r^2 h$. [4]

(c) The semicircle $y = \sqrt{r^2 - x^2}$ for $-r \le x \le r$ is rotated about the x-axis. Show that the volume is $\frac{4}{3}\pi r^3$. [5]

(d) The cone in part (b) has one third of the volume of the cylinder in part (a). Explain where the factor $\frac{1}{3}$ comes from in the integration. [3]

36. When parts goes in a circle. Some integrals return to where they started, and that is what solves them.

(a) Let $I = \displaystyle\int e^x \cos x \, dx$. Apply integration by parts twice, and show that $I = \frac{1}{2}e^x(\sin x + \cos x) + C$. [6]

(b) Explain why choosing $u = e^x$ at the first step and $u = \cos x$ at the second would not have worked. [3]

(c) Find $\displaystyle\int x^2 e^x \, dx$ by applying parts twice, and explain why this one terminates while part (a) does not. [5]

(d) Predict, without calculating, how many rounds of parts $\displaystyle\int x^4 e^x \, dx$ requires. Justify your answer. [2]

37. A reduction formula. Let $I_n = \displaystyle\int_0^{\pi/2} \sin^n x \, dx$ for $n \ge 0$.

(a) Evaluate I_0 and I_1 directly. [3]

(b) Writing $\sin^n x = \sin^{n-1} x \cdot \sin x$ and using integration by parts, show that

$$I_n = \frac{n-1}{n} I_{n-2}, \qquad n \ge 2.$$

[6]

(c) Use the formula to find I_2, I_3, I_4 and I_5 exactly. [4]

(d) Explain why I_n involves π when n is even but not when n is odd. [3]